Unitary Black Hole Evaporation, Quantum Extremal Surfaces, and the Page Curve

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ABSTRACT

We present a self-contained semiclassical formulation of unitary black hole evaporation that couples dynamical horizon collapse and Quantum Extremal Surface (QES) entanglement-wedge reconstruction with a unified set of dimensionless diagnostic invariants. Modeling exponential mode peeling across an ingoing Vaidya horizon, we trace the thermodynamic transition from early-time linear entanglement growth to late-time radiation purification via the Euclidean replica-wormhole saddle exchange. To audit microstate unitarity, horizon blueshift regularization, and holographic information bounds continuously along Kerr–Newman evaporation trajectories, we integrate the Zero-Angular-Momentum Observer (ZAMO) dilation factor (ρ_τ), Weyl conformal purity (χ), decoherence action (Ω), holographic coherence floor (Φ_{floor}), timeline integrity (Φ_τ), surviving coherent cycles ($\mathcal{N}_{eff}$), the Lloyd–Bekenstein gate ratio (η_{Lloyd}), and the L_∞ semiclassical breakdown index (Ξ_{QG}).

Keywords: Black Hole Information Paradox, Quantum Extremal Surfaces, Island Formula, Replica Wormholes, Page Curve, Unitary Evaporation, Dimensionless Invariants

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1. Dynamical Horizon Collapse and Kinematical Redshift

Consider a spherically symmetric collapsing star modeled by the ingoing **Vaidya spacetime metric**:

$$ds^2 = -(1 - 2GM(v) / c^2 r)\, c^2\, dv^2 + 2c\, dv\, dr + r^2\, d\Omega^2$$

where v denotes advanced null time and $M(v)$ is the evolving mass profile.

- **Exponential Mode Peeling and Bogoliubov Coefficients:** Tracing outgoing null rays backward from future null infinity ($\mathscr{I}^+$) through the collapsing center to past null infinity ($\mathscr{I}^-$) relates asymptotic retarded time u to the affine horizon coordinate U via $U = -\kappa^{-1} e^{-\kappa u}$, where κ is the horizon surface gravity. Evaluating the Klein–Gordon inner product between early-time and late-time field modes yields Bogoliubov coefficients satisfying a thermal Planck spectrum at the **Hawking temperature** [1]:

  $$T_H = \hbar\kappa / (2\pi\, k_B\, c)$$

- **ZAMO Dilation Factor (ρ_τ):** Relates proper time intervals along the local Zero-Angular-Momentum Observer frame ($d\tau_{pre}$) to distant observer proper time ($d\tau_{obs}$) via $\rho_\tau = d\tau_{obs} / d\tau_{pre}$ [2, 3]. It transforms local horizon-frame temperature T_{eff} into the Tolman–Ehrenfest redshifted temperature measured at infinity, T_{obs} =

T_{eff} / ρ_τ. Under the stretched-horizon membrane regulator ($N^2 \geq 10^{-6}$) [4], ρ_τ remains strictly bounded ($\rho_\tau \leq 10^3$), preventing unphysical infinite-blueshift divergences near the horizon.

- **Weyl Conformal Purity (χ):** Quantifies static geometric vacuum polarization induced by background tidal curvature ($\mathcal{A}_W = \mathcal{C}_{clamped}^2 / 60\pi^2$) derived from the Seeley–DeWitt a_4 trace anomaly [5]:

$$\chi = 1 / (1 + 60\, \mathcal{A}_W) = 1 / (1 + \mathcal{C}_{clamped}^2 / \pi^2)$$

In flat Minkowski space ($\mathcal{C}_{clamped} = 0$), $\chi = 1$; at a horizon, χ drops below unity (≈ 0.908 for Schwarzschild, ≈ 0.712 for extremal Kerr–Newman), isolating time-independent curvature mode mixing from thermal particle emission.

2. Generalized Entropy and the Quantum Extremal Surface (Island Formula)

In semiclassical gravity, the von Neumann entropy of Hawking radiation (S_{rad}) collected in a distant region *R* diverges near the horizon due to short-distance ultraviolet entanglement across the boundary. Regulating this divergence requires pairing geometric horizon area with renormalized bulk field entropy on a spacelike slice Σ_X.

- **Generalized Entropy Functional (S_{gen}):** Allowing the radiation's entanglement wedge to include a disconnected interior **island** *I* bounded by a codimension-2 surface $\partial I = X$, the generalized entropy is:

$$S_{gen}(X) = Area(X) / (4\, G\, \hbar) + S_{bulk}(R \cup I)$$

- **Extremization and Minimization (The Island Rule):** Varying *X* along orthogonal future-directed null normals (k^μ outward, l^μ inward) until the quantum expansions vanish ($\Theta_k = \Theta_l = 0$) and selecting the global minimum over all extremal saddles gives [6, 7]:

$$S_{rad}(R) = \min_I \{ \text{ext}_{\partial I} [Area(\partial I) / (4\, G\, \hbar) + S_{bulk}(R \cup I)] \}$$

- **Euclidean Replica Wormhole Saddle Exchange:** Evaluating the *n*-sheeted gravitational path integral via the **Replica Trick**, $S_{rad} = -\lim_{n \to 1} (\partial/\partial n) \ln(Z_n / Z_1^n)$, exposes two competing topological saddles:

1. **The Hawking Saddle ($I = \varnothing$, Disconnected Replicas):** Produces linear entanglement growth $S_{rad}^{(Hawking)}(t) \approx (c_{eff} / 12)\, \kappa\, t$, leading to apparent information loss [1, 8].
2. **The Replica Wormhole Saddle ($I \neq \varnothing$, Connected Replicas):** Connects the *n* copies through a wormhole throat branched at ∂I, yielding $S_{rad}^{(Island)}(t) \approx Area(r_+(t)) / (4\, G\, \hbar) = S_{BH}(t)$ [6, 7].

3. Unitary Page Curve Evolution and Microscopic Coherence Bounds

Summing over both saddles in the Euclidean path integral forces the smaller Euclidean action to dominate exponentially, yielding the unitary **Page Curve** [8]:

$$S_{rad}(t) = \min[\, S_{BH,init} - S_{BH}(t),\quad S_{BH}(t)\,]$$

- **Pre-Page Regime ($t < t_{Page}$):** The trivial saddle ($I = \varnothing$) dominates, and $S_{rad}(t)$ rises as outgoing quanta entangle with interior partners.

- **Page Time ($t = t_{Page}$):** When the horizon area shrinks to half its initial value, $S_{rad}^{(Hawking)}(t_{Page}) = S_{BH}(t_{Page})$, triggering the phase transition to the non-trivial Quantum Extremal Surface.
- **Post-Page Regime ($t > t_{Page}$):** Late Hawking quanta purify early radiation, and $S_{rad}(t)$ tracks $S_{BH}(t)$ downward along the Kerr–Newman evaporation trajectory:

$$dM / dt = -\alpha_H f_{KN}^2 / M^2, \quad f_{KN} = 2\chi_{ext} / (1 + \chi_{ext}), \quad \chi_{ext} = \sqrt{(1 - a_*^2 - Q_*^2)}$$

How Custom Invariants Enforce and Audit Microscopic Unitarity

- **Decoherence Action (Ω) and Timeline Integrity (Φ_τ):** Along the evaporating worldline, thermal Hawking scattering (Γ_{th}) and gravitational self-energy dephasing (Γ_{grav}) accumulate dimensionless decoherence action $\Omega = (\Gamma_{th} + \Gamma_{grav})\, t_{pre}$ [9, 10]. Rather than allowing quantum state purity to vanish identically ($Tr(\rho^2) \to 0$), **Timeline Integrity (Φ_τ)** bounds coherence decay from below:

$$\Phi_\tau(\Omega) = \Phi_{floor} + (1 - \Phi_{floor})\, e^{-\Omega}$$

- **Holographic Coherence Floor (Φ_{floor}):** Derived from the Bekenstein–Hawking Hilbert-space dimension $N_{dof} = 1 + \pi \cdot 10^{2\zeta}$ (where $\zeta = \log_{10}(r_+ / \ell_P)$), $\Phi_{floor} = (1 + \pi \cdot 10^{2\zeta})^{-1}$ represents the irreducible purity of a maximally scrambled state inside a causal sphere of radius r_+. As evaporation shrinks the horizon toward the Planck length ($\zeta \to 0$), Φ_{floor} rises smoothly from $\approx 10^{-70}$ to $(1 + \pi)^{-1} \approx 0.2414$, preserving residual microstate correlations.
- **Surviving Coherent Cycles ($\mathcal{N}_{eff}$) and Lloyd–Bekenstein Ratio (η_{Lloyd}):** Weighting total emitted wave oscillations ($\mathcal{N}_\varphi = f_{local}\, t_{pre}$) by Φ_τ yields $\mathcal{N}_{eff} = \mathcal{N}_\varphi \cdot \Phi_\tau$. Comparing $\mathcal{N}_{eff}$ against the shrinking horizon's holographic entropy capacity defines the **Lloyd–Bekenstein Gate Ratio** [11, 12]:

$$\eta_{Lloyd} = \mathcal{N}_{eff} / (S_{BH} / k_B) \leq 1$$

- **Semiclassical Breakdown Index (Ξ_{QG}):** As $M(t) \to M_P$ and $r_+ \to \ell_P$, the minimal-length ratio $\Pi_1 = \ell_P / L$ and thermal-foam ratio $\Pi_4 = T_H / T_P$ approach unity. The L_∞ supremum monitor $\Xi_{QG} = \max(\Pi_1, \Pi_2, \Pi_3, \Pi_4) \to 1$ signals the exact breakdown of smooth semiclassical geometry, halting evaporation at a stable **Planck-mass relic (M_P)**.

4. Operational Interpretation and Diagnostic Summary

Symbol	Name	Definition	Operational Role in Unitary Evaporation
ρ_τ	ZAMO Dilation Factor	$(\gamma_{pre}/\sqrt{N^2_{pre}}) / (\gamma_{obs}/\sqrt{(1-\mathcal{C}_{obs})})$	Converts local horizon evaporation rates and temperatures into distant observer frames without coordinate singularities.
χ	Weyl Conformal Purity	$(1 + \mathcal{C}_{clamped}{}^2 / \pi^2)^{-1}$	Isolates static curvature-induced vacuum polarization from dynamical thermal Hawking radiation.
Ω	Decoherence Action	$(\Gamma_{th} + \Gamma_{grav})\, t_{pre}$	Quantifies accumulated thermal and gravitational phase scattering along the evaporating worldline.
Φ_{floor}	Holographic Coherence Floor	$(1 + \pi \cdot 10^{2\zeta})^{-1}$	Sets the irreducible inverse Hilbert-space purity bound preventing complete information loss.
Φ_τ	Timeline Integrity	$\Phi_{floor} + (1 - \Phi_{floor})e^{-\Omega}$	Tracks surviving quantum phase coherence from initial pure state (1) to scrambled equilibrium (Φ_{floor}).
$\mathcal{N}_{eff}$	Surviving Coherent Cycles	$(f_{local}\, t_{pre})\, \Phi_\tau$	Counts phase-coherent wave oscillations available for unitary information transfer.
η_{Lloyd}	Lloyd–Bekenstein Ratio	$\mathcal{N}_{eff} / (S_{BH} / k_B)$	Audits that coherent information emission obeys the instantaneous Bekenstein–Hawking bit limit ($\eta_{Lloyd} \leq 1$).
Ξ_{QG}	Semiclassical Breakdown Index	$\max(\Pi_1, \Pi_2, \Pi_3, \Pi_4)$	Halts semiclassical extrapolation at the Planck-mass relic boundary ($M \to M_P$) when $\Xi_{QG} \to 1$.

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